

# Day 17: DAX Design Patterns & Dynamic Metrics

## 📌 Definition to Remember

**DAX Design Patterns** are standardized, battle-tested architectural templates for solving recurring analytical challenges such as dynamic metric switching, customer segmentation (RFM), and 80/20 Pareto analysis.

### 1. Dynamic Metric Selection via Disconnected Tables / Field Parameters

Allow end users to toggle visual metrics (Sales, Profit, Quantity, Margin) from a single slicer:

```
Dynamic Metric Value =  
SWITCH(  
    SELECTEDVALUE(MetricSelection[MetricName]),  
    "Total Sales", [Total Sales],  
    "Total Profit", [Total Profit],  
    "Order Volume", [Total Quantity],  
    "Profit Margin", [Profit Margin %],  
    [Total Sales] // Default fallback  
)
```

### 2. Dynamic Customer Segmentation

Classifying customers dynamically without hardcoding static columns:

```
Customer Segment =  
VAR CustSales = [Total Sales]  
RETURN  
    SWITCH(  
        TRUE(),  
        CustSales >= 100000, "Tier 1: High Value",  
        CustSales >= 25000, "Tier 2: Medium Value",  
        "Tier 3: Low Value"  
    )
```

### 3. Pareto 80/20 Analysis Pattern

Calculating cumulative contribution % to identify the top 20% of products driving 80% of total revenue:

```
Cumulative Sales % =  
VAR CurrentSales = [Total Sales]  
VAR TotalRevenue = CALCULATE([Total Sales], ALLSELECTED(Dim_Product))  
VAR CumulativeSum =  
    CALCULATE(  
        [Total Sales],  
        FILTER(  
            ALLSELECTED(Dim_Product),  
            [Total Sales] >= CurrentSales  
        )  
    )  
RETURN  
    DIVIDE(CumulativeSum, TotalRevenue, 0)
```

### ★ Must-Master Skills & Interview Points

1. Creating Field Parameters in Power BI Desktop for seamless visual metric switching.
2. How dynamic segmentation responds in real time to date and region slicers.
3. Implementing Pareto 80/20 curves on combo charts.

### ⚡ Quick Recall

Field Parameters (Dynamic Metrics) + Dynamic Segmentation + Cumulative Pareto 80/20 Analysis